COMPUTER APPLICATION TEST

Class – 7

Full marks – 50 (pass 60%)

(Write the option with the answer only don't write the questions)

Time - (One and a half Hours)

(Marks X Questions)

SECTION A: Theoretical Foundations (30 Marks Total)

General Instructions for Section A:

- Answer all questions.
- Each question carries exactly 1 mark.
- For Multiple Choice Questions, select the single most appropriate option.

Q1. Multiple Choice Questions (15 Marks — 1 Mark Each)

1. What does the input() function return by default when a user types a number? a) int b) float c) str d) bool
2. Which built-in method adds a single element to the *end* of a list? a) .insert() b) .append() c) .add() d) .update()
3. In Python, what is the mathematical output of 15 // 4? a) 3.75 b) 4 c) 3 d) 15
4. Which of the following data structures is unordered, unindexed, and does not allow duplicate values? a) List b) Tuple c) Dictionary d) Set
5. What is the correct syntax for a shorthand if/else (ternary) statement in Python? a) if condition: x else: y b) x if condition else y c) x else y if condition d) if (condition, x, y)
6. If text = "Hello World", what does text[::-1] evaluate to? a) Prints only the first word b) Reverses the entire string c) Throws an IndexError d) Removes all white spaces
7. In a Python 3.10+ match-case statement, what symbol is used to combine multiple cases (acting as the OR operator)? a) || b) or c) | d) &
8. How do you safely retrieve a value from a dictionary without risking a KeyError if the key does not exist? a) del dict["key"] b) dict.get("key", "default") c) dict.remove("key") d) dict.popitem()
9. Which keyword is used inside a loop to completely skip the rest of the current iteration and jump back to the top? a) break b) continue c) pass d) return
10. What does the is identity operator strictly evaluate in Python? a) If two variables have the exact same data value b) If two variables point to the exact same object in memory c) If two variables are of the same data type d) If a variable is declared as public
11. Which list method merges all elements from another iterable (like a second list) into the current list? a) .extend() b) .join() c) .append() d) .combine()
12. If a = True and b = False, what is the boolean result of a and not b? a) True b) False c) None d) 0
13. Which string method breaks a string into a list of substrings based on a specified separator? a) .slice() b) .divide() c) .split() d) .separate()
14. What does the .strip() string method do? a) Removes all vowels from the string b) Removes leading and trailing whitespaces from the string c) Empties all characters from a list d) Converts the string to lowercase
15. If a while loop finishes its condition normally without hitting a break statement, which block of code will execute immediately after? a) finally: b) else: c) pass: d) except:

Q2. Fill in the Blanks (10 Marks — 1 Mark Each)

1. A tuple is ________, meaning its internal elements cannot be added, removed, or changed after creation.
2. The ________ built-in function is used to find the total number of items in a dictionary, list, set, or string.
3. The arithmetic operator is used in Python to perform ________ (exponentiation/power) calculations.
4. To combine a list of strings into a single string separated by spaces or hyphens, you use the string ________ method.
5. In Python's boolean evaluation, an empty string "" or the number 0 automatically evaluates to ________.
6. The set operator & (or .intersection()) is used to find the ________ elements shared between two sets.
7. To extract only the keys from a dictionary as a list-like view, you use the ________ method.
8. The ________ keyword acts as a null statement placeholder when syntax requires code but no action is needed.
9. Using the list slice mylist[1:4] will extract items starting from index 1 up to, but not including, index ________.
10. If you assign x = 10, the shorthand compound assignment statement to add 5 directly to x is written as ________.

Q3. True or False (5 Marks — 1 Mark Each)

1. A Python dictionary can contain duplicate key names, and it will keep all of them in memory. *(True / False)*
2. In a for loop written as for i in range(1, 10, 2):, the printed sequence of numbers will be 1, 3, 5, 7, and 9. *(True / False)*
3. The .pop() method on a List removes an item by its index position, whereas the .pop() method on a Set removes an arbitrary/random item. *(True / False)*
4. You can use the in membership operator inside an if statement to check whether a specific substring or character exists inside a string. *(True / False)*
5. The int() casting function can successfully convert a decimal string like "45.8" directly into the integer 45 without throwing an error. *(True / False)*

<u>SECTION B</u>: Practical Industry Scenarios (70 Marks Total)

General Instructions for Section B:

- Solve all 5 real-world problems below.
- Write top-to-bottom procedural scripts. Do **not** use custom functions (def) or global variables.
- Follow all explicit steps, formulas, and structural hints carefully.

Q4. The Supermarket Billing System (15 Marks)

Context: You are writing the backend script for a supermarket self-checkout terminal. It must loop infinitely to scan items until the customer types a specific exit command.

The Task:

1. Create an empty dictionary called cart = {} and a variable subtotal = 0. *(2 Marks)*
2. Write a while True: loop. *(1 Mark)*
3. Inside the loop, use input() to ask: "Enter item name (or type 'pay' to finish): ". *(2 Marks)*
4. If the input is "pay", use the break keyword to exit the loop immediately. *(2 Marks)*
5. Otherwise, use input() to ask for the "Price: ". Cast this price directly to a float. *(2 Marks)*
6. Add the item as the key and the price as the value in the cart dictionary, and add the price to your subtotal. *(2 Marks)*
7. **After the loop breaks**, calculate the 18% GST tax and the final total bill using the formulas below: *(2 Marks)*
   - **GST Formula:** gst = subtotal * 0.18
   - **Final Total Formula:** final_total = subtotal + gst
8. Print a formatted receipt using an f-string showing the Subtotal, GST, and Final Total rounded to exactly 2 decimal places (:.2f). *(2 Marks)*

- **Hint:** Always check for your break condition ("pay") BEFORE asking for the item price, otherwise the word "pay" will trigger a casting error or end up stored in your cart dictionary!

Q5. SGPA Aggregator & Grade Matcher (15 Marks)

Context: You are writing a script to calculate a student's Semester Grade Point Average (SGPA) across 4 subjects based on their letter grades using Python 3.10's modern match-case statement.

The Task:

1. Use input() to ask for the student's name. *(1 Mark)*
2. Create two variables: total_grade_points = 0 and total_credits = 12 *(Assume each of the 4 subjects is worth 3 credits: 4 × 3 = 12)*. *(1 Mark)*
3. Write a for loop that runs exactly 4 times using for i in range(4):. *(1 Mark)*
4. Inside the loop, use input() to ask the user: "Enter letter grade for Subject (O, E, A, B, C, F): ". Use .upper() to standardize the input. *(2 Marks)*
5. Write a match-case block for the entered grade: *(6 Marks)*
    - case "O": Add 30 to total_grade_points *(10 points × 3 credits)*.
    - case "E": Add 27 to total_grade_points *(9 points × 3 credits)*.
    - case "A": Add 24 to total_grade_points *(8 points × 3 credits)*.
    - case "B": Add 21 to total_grade_points *(7 points × 3 credits)*.
    - case "C": Add 18 to total_grade_points *(6 points × 3 credits)*.
    - case "F": Add 0 to total_grade_points.
    - case _: Print "Invalid Grade" and use continue to skip the rest of the loop.
6. Outside the loop, calculate the final SGPA using the formula below: *(2 Marks)*
    - **SGPA Formula:** sgpa = total_grade_points / total_credits
7. Print a final result string: "[Name]'s final SGPA is [sgpa]". *(2 Marks)*

Q6. Bank Transaction Ledger & Strings (15 Marks)

Context: You exported your digital bank statement, but it generated as a list of raw strings mixing the transaction type and the amount together with a hyphen. You need to parse these strings to calculate your final bank balance and track basic metrics.

Starter Data:

Python

```
monday_class = {"Amit", "Rohan", "Priya", "Sneha"}

friday_class = {"Priya", "Karan", "Amit", "Vikram"}
```

The Task:

1. Create a variable bank_balance = 0. *(1 Mark)*
2. Create a dictionary to track metrics: metrics = {"Total_Deposits": 0, "Total_Withdrawals": 0}. *(2 Marks)*
3. Write a for loop to iterate through the transactions list. *(1 Mark)*
4. Inside the loop, use .split("-") to break the string into two parts. Extract the action (index 0) and the amount (index 1). Cast the extracted amount directly to an int. *(3 Marks)*
5. Write an if/elif block to process the transaction: *(5 Marks)*
    - If the action is "Deposit", add the amount to bank_balance AND add the amount to metrics["Total_Deposits"].
    - If the action is "Withdraw", subtract the amount from bank_balance AND add the amount to metrics["Total_Withdrawals"].
6. After the loop finishes, print the final bank_balance and the complete metrics dictionary clearly. *(3 Marks)*

- **Hint:** Remember that splitting "Deposit-5000" gives you a list of strings: ["Deposit", "5000"]. You must extract index 1 and cast it to an integer before performing any mathematical addition or subtraction!

Q7. The Cloud Server Log Filter (15 Marks)

Context: Your cloud application generated a batch of raw system logs. You need to write a script that isolates all the log entries flagged as "ERROR", extracts their error messages into a list, and records which unique IP addresses are causing the problems.

Starter Data:

Python

```
logs = [

    "192.168.0.1 | INFO | Boot successful",

    "10.0.0.5 | ERROR | Timeout connection",

    "192.168.0.1 | INFO | User logged in",

    "172.16.0.8 | ERROR | Database locked",

    "10.0.0.5 | ERROR | Retry failed"

]
```

The Task:

1. Create an empty list called error_messages and an empty set called trouble_ips = set(). *(2 Marks)*
2. Write a for loop to iterate through the logs list. *(1 Mark)*
3. Inside the loop, use .split(" | ") to break the string into three distinct parts: IP Address, Log Type, and Message. *(3 Marks)*
4. Write an if statement to check if the Log Type exactly equals "ERROR". *(2 Marks)*
5. If the log is an Error, perform two actions inside the if block: *(4 Marks)*
    - Use .append() to save the Message part to the error_messages list.
    - Use .add() to save the IP Address part to your trouble_ips set *(The set will automatically filter out any duplicate IP addresses)*.
6. If the Log Type is "INFO", use the pass keyword to ignore it. *(1 Mark)*
7. After the loop completes, print the trouble_ips set and the error_messages list on separate lines. *(2 Marks)*

Q8. Gym Membership Venn Diagram (10 Marks)

Context: A local fitness center is analyzing user attendance using Python Set Theory to determine which members are consistent attendees and which members have irregular workout schedules.

Starter Data:

Python

```
monday_class = {"Amit", "Rohan", "Priya", "Sneha"}

friday_class = {"Priya", "Karan", "Amit", "Vikram"}
```

The Task:

1. Use the correct Set operator to find the members who attended **both** classes (the intersection). Save this resulting set to a variable called consistent_members. *(3 Marks)*
2. Use the correct Set operator to find the members who attended **only one** of the classes, but not both (the symmetric difference). Save this resulting set to a variable called irregular_members. *(3 Marks)*
3. Cast irregular_members into a standard Python list(), then use .sort() to arrange the names in alphabetical order. *(2 Marks)*
4. Print the consistent_members set, and print the sorted irregular_members list clearly. *(2 Marks)*

- **Hint 1:** To find the intersection (elements shared by both sets), use the & operator or .intersection() method.
- **Hint 2:** To find the symmetric difference (elements present in one set or the other, but strictly not both), use the ^ operator or .symmetric_difference() method.